OFFICIAL COLLEGE PROJECT ABSTRACT · PROJECT 28 OF 55

AI-BASED AIR QUALITY FORECASTING AND POLLUTION LEVEL PREDICTION USING TIME-SERIES DEEP LEARNING

Domain: Environmental Modeling & Time-Series AI | **Module:** project_28_air_quality_forecast

OFFICIAL ABSTRACT

"AI-Based Air Quality Forecasting and Pollution Level Prediction Using Time-Series Deep Learning is a comprehensive and crucial project aimed at addressing unexpected environmental hazards, public health risks, and air pollution inefficiencies in modern urban and industrial environments. As industrialization and urbanization accelerate, cities increasingly rely on interconnected monitoring networks that continuously generate complex environmental metrics, making them prime targets for sudden hazardous pollution spikes if air quality degradation is not proactively identified. Traditional monitoring methods often rely on scheduled or reactive reporting, which can be time-consuming, resource-intensive, and fundamentally inefficient against rapidly developing, complex atmospheric anomalies and subtle pollutant shifts. To address these persistent challenges and limitations, this study proposes an advanced, multi-layered approach combining cutting-edge deep learning techniques with intelligent time-series analysis algorithms for efficient, automated, and real-time environmental monitoring and pollution level forecasting. The system utilizes specialized deep learning architectures to continuously capture and extract key behavioral features from extensive temporal sensor streams, such as fluctuating particulate matter levels, gas concentrations, meteorological parameters, and operational wind speed deviations. Once these underlying structural patterns are mapped, the extracted features are fed into classification and forecasting algorithms designed to enhance decision-making, optimize computational overhead, improve prediction accuracy, and drastically reduce false alarm and misclassification rates. By profiling what constitutes normal, routine atmosphere for each distinct geographic zone, the model establishes a dynamic baseline that adapts to changing climatic and traffic load requirements over time. The proposed model is rigorously trained, validated, and tested on robust datasets encompassing diverse environmental sensor

> logs, operational cycles, and simulated pollution scenarios. " "AI-BASED SIGN LANGUAGE TRANSLATION WITH REAL-TIME SPEECH AND TEXT GENERATION

📌 Project Summary & System Specifications

Project Title	AI-BASED AIR QUALITY FORECASTING AND POLLUTION LEVEL PREDICTION USING TIME-SERIES DEEP LEARNING
Specialization / Category	Environmental Modeling & Time-Series AI
Target Technology Stack	Python (Flask), PyTorch / TensorFlow / Scikit-Learn, Modern Web UI
Default Backend Port	`:5028`
Local Implementation	`project_28_air_quality_forecast`